CEN-504

B.Tech. (Computer Engineering) V Semester Examination, 2022

Object Oriented Programming

Roll No. 20BCS021

Paper No. CEN-504

Time: Three Hours

Maximum Marks: 60

Note: Attempt any two parts from each question.
Make suitable assumptions wherever necessary.

Q.1 CO1	(a)	Write a program in C++ for the bookshop using appropriate member function and constructors for the following scenario: A bookshop keeps track of the books that are available for purchase. The list includes information such as the author's name, title, cost, publication, and stock position. When a customer requests a book, the salesperson enters the title and author, and the system scans the list to see if it is available. If it is not, a suitable notice is displayed. If it is, the system displays the book information and requests the number of copies needed. If the requested copies are in stock, the total cost is displayed; otherwise, the message "Sorry! These many copies are not in stock" is displayed.	6
	(b)	In object-oriented programming, we can create a class that helps to find the solution to a mathematical equation. One of the equations we often need to solve in algebra is the quadratic equation as shown below: **ax2 + bx + c = 0** The roots of this equation are: **x1 = (-b + sqrt (b2 - 4 * a * c))/2*a** **x2 = (-b - sqrt (b2 - 4 * a * c))/2*a** The phrase inside the parentheses is called the discriminant. If the value of the discriminant is positive, the equation has two roots. If it is zero, the equation has one root. If it is negative, the equation has no roots. Write a program in C++ program that finds the root of a quadratic equation when the coefficients a, b, and c are given.	6
	(c)	Define a class Matrix with an integer array of 3X3 as a data member. Define a friend function which multiplies two matrix objects and returns resultant matrix object.	6
Q.2 CO2	(a)	Declare and define a class Set representing a set of integers. A set is a collection of data without repetition or ordering. The class should have only private data members: a pointer to a dynamically allocated array of integers and an integer that holds the size of the set. The following methods are required to implement: ➢ A constructor to create an empty set. ➢ A function to add an element to the set (overload the += operator). ➢ A function to remove an element from the set (overload the -= operator). ➢ A binary friend function to get the union of two sets (overload the + operator). ➢ A binary friend function to get the intersection of two sets (overload the * operator). ➢ A function to print the contents of a set	6

	(b)	What is the benefits of operator overloading. Write a program in C++ to overload new and delete operators to manipulate objects of the Student class. The Student class must contain data members such as char *name, int roll_no, int branch, etc. The overloaded new and delete operators must allocate memory for the Student class object and its data members.	
	(c)	Discuss conversion between object of different classes. Write a C++ program that perform conversion between degrees and radian that has following relation: ➢ **Angle in Radian = Angle in Degree * PI /180.0** ➢ **Angle in Degree = Angle in Radian * 180.0 / PI, where PI = 22/7** The conversion routine should be implemented in the following scenario: ➢ When conversion routine in source object. ➢ When conversion routine in destination object.	6
Q.3 CO3	(a)	Write an interactive C++ program to create a graphic class hierarchy. Create an abstract base class called Figure and derive two classes Close and Open from that. Declare two more classes called Polygon and Ellipse using the Close class. Create derived classes Line and Polyline from the Open class. Define three objects (triangle, rectangle, and pentagon) of the class Polygon. All classes must have appropriate member functions including constructors and destructors.	6
	(b)	When does ambiguity arise in multiple inheritance? How can one resolve it? What are virtual base classes? What is their significance? Demonstrate the concept of virtual base class with suitable program.	6
	(c)	(i) What are pure virtual functions? What are the rules that need to be kept in mind while deciding virtual functions?	3
		(ii) Write a C++ program to create a base class called shape. Use this class to store two double type values that could be used to compute the area of figures. Derive two specific classes called triangle and rectangle from the base-shape. Add to the base class, a member function get_data() to initialize base-class data members and another member function display_area() to compute and display the area of figures. Make display_area() as a virtual function and redefine this function in derived classes to suit their requirements.	3
Q.4 CO4	(a)	Define a class called Fuel_Monitor that will be used to check the amount of fuel that is left over in a vehicle after travelling a certain distance. The class should have an instance variable tank_capacity to store initial size of the tank and efficiency to store initial efficiency of the vehicle. Also, set the variable fuel_in_tank to zero that is used to store initial fuel in the tank. Include a method that returns ini_tank_size and fuel_in_tank. Include an method add_fuel that calculates how much fuel can be filled depending upon the fuel already in the tank and the capacity of the tank. Also, include a method drive_distance that returns how much distance can be travelled with the fuel available in the tank with efficiency provided. Embed your class in a test program with appropriate use of public, private and protected access specifier.	6

	(b)	(i) What are the benefits of using packages? Write down the steps in creating a package and using it in a Java program with an example. (ii) Discuss the criteria's for applying the concept of interface and abstract class in the context of Java programming. Take an appropriate programming example to support your criteria' s.	3 3
	(c)	Define an interface using JAVA that contains a method to calculate the perimeter of an object. Define two classes circle and Rectangle with suitable fields and methods. Implement the interface "perimeter" in these classes. Write the appropriate main () method to create object of each class and test all the methods.	6
Q.5 CO5	(a)	Which keywords are used to handle exceptions in Java? Demonstrate the use of these keywords with the help of small piece of codes.	6
	(b)	(i) Differentiate between checked exception and unchecked exception with the help of Java code. (ii) An University has applied promotion criteria for students. According to criteria a student cannot promote to next academic year if he has less than 4.5 CGPA. A developer is trying to implement this situation using exception handling in JAVA. Write a correct Java code to help him.	3 3
	(c)	(i) What is the difference between finally and finalize blocks? Demonstrate with the help of Java program. (ii) What is a Thread? Which Interface must be implemented by all threads? What are the ways to create a thread? How will you start and stop a thread? How do we set priorities for threads?	3 3